

there's how it aids in identifying a suitable research problem

1. Identifying Gaps in Knowledge:

A literature review reveals areas where research is lacking or inconclusive, helping researchers pinpoint unanswered questions or unexplored aspects that warrant investigation.

2. Refine Research Focus:

By understanding what has already been studied, researchers can narrow down broad topics into specific manageable research problems.

3. Provide context:

It situates the research within the existing body of knowledge, ensuring the problem is relevant and significant to the field.

4. Avoids Duplication:

Reviewing prior work helps researchers avoid redundant studies, ensuring their work contributes something new.

b. Informs Methodology

Insights from previous studies guide the selection of appropriate methods and approaches for addressing the research problem

b. Justifies Research Need:

A lit. rev establishes the importance of the research problem by demonstrating its relevance to current issues or debates in the field.

In this summary, a literature review is foundational in identifying a suitable research problem by providing a

comprehensive understanding of what is unknown, and where new research can make a meaningful contribution

- a) choose a topic of research from your specialization and formulate a hypothesis of the research. Describe from the above perspective in brief how the research approach and Methodology will vary in case of
a) Analytical research vs experimental research
b) Explanatory research vs exploratory research

Research Topic statement.

Drones in Agricultural Applications:

Enhancing Precision Farming

Formulated hypothesis:

Enhancing precision farming using drones equipped with multispectral imaging and AI driven data.

The Integration of drones in agricultural applications enhances precise farming.

Research Type	Key Difference in Approach	Key difference in Methodology
Analytical Research	Analytical: Deductive and Theoretical focusing on existing data and models without real-world testing	Analytical: Literature review, mathematical modelling simulations
Experimental Research	Experimental: Inductive and empirical emphasizing hands-on- trials and direct data Collection.	Experimental: Drones, deployment on farms, data collections over seasons statistical analysis of predictions.

Explanatory Research

VS

Exploratory Research

Explanatory is structured and

hypothesis driven, aiming to establish causal relationship based on prior evidence

Exploratory:

open-ended and inductive

seeking to uncover new

patterns without rigid

assumptions

Explanatory: Regression analysis

controlled comparisons,

machine learning

for causal modelling

Exploratory

Farmer interviews

case studies,

Pilot tests in varied

conditions for discovery

7) CNT's platforms in oncology. → 5 questionnaires

Nanotechnology in Cancer Treatment

1) What does best describes cancer based on its key characteristics?

- A single disease caused by viral infections
- A group of disease involving uncontro

10.10.2015

Design of Experiments (DOE)

* DOE is invented by Sir Ronald A. Fisher in 1920's and 1930's

* The brief history of design of experiments is summarized in the following 4 eras:

1. The agricultural origin: 1918 to 1940

— Profound impact on agricultural sciences

— Development of factorial design and ANOVA

2. The first industrial era 1951 to late 1970's

— Applications in the chemical and process industries

3. The second industrial era late 1970's to 1990

— Quality improvement initiatives in many companies

— Total quality management (TQM) and continuous quality improvement (CQI) were important ideas and became management goals

Factor

* Controllable variables that are deliberately manipulated to determine the individual and joint effects on the responses), OR Factor are those quantities that affect the outcome of an experiment, Eg: heat source, lid, shape of pot, etc.

* A variable or attribute which influences or is suspected of influencing the characteristic being investigated.

* Two types — Quantitative, Qualitative

Level

Treatment

* one set of levels of all factors employed in a given experimental trial.

* In the case of single factor experiment, each level of the factor is a treatment

Run or trial

* Running the exp. to get o/p.

Experimental Unit

→ Facility with which an exp trial is conducted such as samples of materials, person, animal, plant, etc.

Response

- Result/output obtained from a trial of an experiment.
- also called dependent variable.

Effect

- Effect of a factor is the change in response due to change in the level of the factor.

Experimental Error

* Diff b/w expected outcome and actual outcome.

Need of DOE Methodology

→ Experimental design methods are ahoit of fundamental importance in engg. design activities where new products are developed and existing ones improved.

* Evaluation and comparison of basic design configurations.

* Evaluation of material alternatives.

* selection of design parameters so that the product will work well under a wide variety of field conditions, that is the product is robust

* determination of key product design parameters that impact

Full Factorial design, Basic concepts of 2^2 , 2^3 and 2^k designs.

* It is widely accepted that the most commonly used experimental designs in manufacturing companies are full and fractional factorial designs at 2-level and 3 levels.

$2^n \rightarrow n$ indicates trials and also parameters

1 Factor $2^1 = 2$ Runs

A $\rightarrow$ parameter

Low High

A

2 Factor $2^2 = 4$ Runs

A, B $\rightarrow$ Parameters

A B
-1 +1
+1 -1
-1 -1
+1 +1

3 Factor $2^3 = 8$ Run

A, B, C $\rightarrow$ Parameters

Full Factorial / Partial Factorial Design

Total parameters n .

— Now considering only 2 factors are considered, then it is called partial factorial design.

— Considering all n parameters, it is called as full factorial design.

A 2^2 design is a simple factorial design where there are two factors, each at two levels. It's often used in exp. design to evaluate the

Analysing the design.

- The main effect of each other factor
- The interaction effect between Temperature & Pressure
- ANOVA table

1	1
1	1
1	1
1	1

- AB CD E
- AB AC AD AE BA BC BD BE
- CA CB CD CE DA DB DC
- DE EA EB EC ED
- ABC ABD ABE ACD ACE ADE

Univ Qn pattern

5x20=100.

out 7, 5 have to be

Types of experiment design

The following designs of experiments will be usually followed.

- 1) Completely Randomised Design (CRD)
- 2) Randomised complete block design (RCBD)
- 3) Latin square design (LSD)
- 4) Factorial design or experiment.
 - 5) confounding
 - 6) split & strip plot design.

* The factorial design will now be called an $p \times q$ exp.

* The total no. of treatments in a factorial exp. is the product of the levels in each factor, in two square

factorial example, the no. of treatments $2 \times 2 = 4$.

$2^3 = 8$. The no. of treatments increases rapidly with an

increase in the no. of factor or an increase in the levels in each factor

* For a factorial experiment involving 5 varieties or types

the total no. of treatments would be $5 \times 4 \times 3 = 60$.

$$1 - \frac{1}{n}$$

2^2 Factorial Experiment

* A special set of factorial experiment consists of experiments in which all factors have 2 levels. Such experiments are referred to generally as 2^n factorials.

* If there are four factors each at two levels the experiment is known as $2 \times 2 \times 2 \times 2$ or 2^4 factorial.

(ii) $L^n \rightarrow$ where L levels
 n factors.

* The calculation of the sum of square is as follows:
correction factor (CF) = $\frac{(GT)^2}{n}$

GT $\rightarrow$ grand total

$n \rightarrow$ total no. of observations.

$$\text{Total sum of squares} = \sum x^2 - CF$$

Replication sum of squares RFF =

$$\text{Treatment sum of square (TSS)} = \frac{(T_1^2 + T_2^2 + \dots + T_m^2)}{n} - CF$$

(or)

$$= \frac{1}{n} \sum T^2 - CF.$$

Residual sum of square = error sum of squares =
 total sum of square - replication sum of square - TSS

ANOVA TABLE

Source of variation

Degree of freedom

Sum of Squares

Mean Sum of squares

Factor

RMS/EMS

Replication

$r - 1$

RSS

$RSS / (r - 1)$

Treatments

$t - 1$

TSS

$TSS / (t - 1)$

TMS/EMS

Error

$(r - 1)(t - 1)$

ESS

$ESS / ((r - 1)(t - 1))$

Total

$rt - 1$

CRD

completeness + uniformity = orthogonality

* Treatments

- assigned completely random.

Fig. 1) consider 30 pots (1-30), and apply 3 treatment (A, B, C) to the pots. Each pot has received a single treatment which is assigned randomly. (homogeneous)

It is shown in table as follows:

A	B	B	B	C
B	C	A	C	A
A	A	B	C	B
A	B	C	A	C
B	A	A	B	B
C	B	C	C	A

* Source of variation

— Treatment - source of variation of interest

— Experimental Error - factors beyond control

$$* \text{variability}_{\text{total}} = \text{Variability}_{\text{treatment}} + \text{Variability}_{\text{Exp. Error}}$$

$$\text{variability}_{\text{to}} = \text{Variability}_t + \text{Variability}_e$$

Eg: 1. Contd...

Hypothesis: Effect of the fertilizer dose affects

the weight of the plant.

Design: CRD

variables: treatment, Exp. Error.

* Fresh weight (g) of plants under different doses of fertilizer (treatments A-C).

A	B	B	B	C		
32	23	48	17	42	1	
C	C	A	C	A	2	
31	41	30	31	39	2	
A	A	B	C	B	7	
32	32	29	19	34	8	
A	B	C	A	C	P	
46	17	27	41	41	01	
B	A	A	B	B		
47	33	43	37	25		
C	B	C	C	A		
33	39	32	27	32		

* Grouping the above table based on treatment in homogeneous manner.

$$\frac{S}{n} = \frac{C}{2}$$

relates to $n = 0$ greater

relates to $n = 8$

relates to $n = 10$

ANOVA table

Analysis of

	A	B	C
1	32	47	31
2	32	23	33
3	46	17	41
4	32	39	27
5	33	48	32
6	30	29	31
7	43	17	19
8	41	37	27
9	39	34	42
10	32	25	41

calculations:

step 1 :

use f test and t test formula.

$$F = \frac{s_1^2}{s_2^2} = \frac{\text{largest variance}}{\text{smallest variance}}$$

$$s^2 = \frac{\sum x^2}{n-1}$$

where n = no. of samples.

x = weight.

$$\text{variance} \Rightarrow V = n-1,$$

	A	$x_1 = x_1 - \bar{x}_1$	x_1^2	B	$x_2 = x_2 - \bar{x}_2$	x_2^2	C	$x_3 = x_3 - \bar{x}_3$	x_3^2
1	32	-4	16	47	15.4	237.16	31	-1.4	1.96
2	32	-4	16	23	-8.6	73.96	33	0.6	0.36
3	46	10	100	17	-14.6	213.16	41	8.6	73.96
4	32	-4	16	39	7.4	54.76	27	-5.4	29.16
5	33	-3	9	48	16.4	268.96	32	-0.4	0.16
6	30	-6	36	29	-2.6	6.76	31	-1.4	1.96
7	43	7	49	17	-14.6	213.16	19	-13.4	179.56
8	41	5	25	37	5.4	29.16	27	-5.4	29.16
9	39	3	9	34	2.4	5.76	42	9.6	92.16
10	32	-4	16	25	-6.6	43.56	47	8.6	73.96
Avg	$\bar{x}$	36	292	31.6		1141.3	32.4		482.4

$$s_1^2 = \frac{292}{10-1}$$

$$s_2^2 = \frac{1141.3}{10-1}$$

$$s_3^2 = \frac{482.4}{10-1}$$

$$s_1^2 = 32.44$$

$$s_2^2 = 126.778$$

$$s_3^2 = 53.6$$

$$s_1^2 = 1052.613$$

$$s_2^2 = 16072.60$$

$$s_3^2 = 2872.96$$

$$\text{Total Treatment} = 360 + 316 + 324$$

$$N = 1000$$

$$\text{Total } A^2 = 13252 \quad (32^2 + 32^2 + 46^2 + 32^2 + 33^2 + 30^2 + 43^2 + 41^2 + 39^2 + 32^2)$$

$$\text{Total } B^2 = 11132 \quad (47^2 + 23^2 + 17^2 + 39^2 + 48^2 + 29^2 + 17^2 + 37^2 + 34^2 + 25^2)$$

$$\text{Total } C^2 = 10980 \quad (31^2 + 33^2 + 41^2 + 27^2 + 32^2 + 31^2 + 19^2 + 27^2 + 42^2 + 47^2)$$

Descriptive statistics

Treatment A

Treatment B

Treatment C

Total

(Sample 1)

(Sample 2)

(Sample 3)

$$n=10$$

$$n=10$$

$$n=10$$

$$30$$

$$\bar{x}_1 = 36$$

$$\bar{x}_2 = 31.6$$

$$\bar{x}_3 = 32.4$$

$$36$$

$$s^2 = 32.44$$

$$s^2 = 127.38$$

$$s^2 = 53.6$$

$$36$$

$$\sum x = 360$$

$$\sum x = 316$$

$$\sum x = 324$$

$$\sum x = 1000$$

$$(\sum x)^2 = 129600$$

$$(\sum x)^2 = 99856$$

$$(\sum x)^2 = 104916$$

$$36$$

$$\sum x^2 = 13252$$

$$\sum x^2 = 11132$$

$$\sum x^2 = 10980$$

$$\sum x^2 = 35364$$

$$s^2 = 32.44$$

$$s^2 = 127.38$$

$$s^2 = 53.6$$

Weight of plant = f_i (fertilizer dose)

Step 2:

Find F test value

$$F_{max} = \frac{\text{largest variance}}{\text{Smallest variance}} = \frac{V_1}{V_2}$$

$$= \frac{127.38}{32.44}$$

$$F_{max} = 3.925$$

$\alpha = \frac{\text{no. of samples/treatment}}{f_{22}}$ level of significance is assumed to be 5%.

$TS = \left(\frac{\sum (x_i^2)}{n} \right) + \left(\frac{\sum (x_i^2)}{n} \right) + \left(\frac{\sum (x_i^2)}{n} \right)$ * Note: If it is given in sum

use the table in appendix of

take it otherwise use 5%.

RM by Cochran's book: 01 + 0.28 PP + 0.03 PC1 =

V_1 in row, V_2 in column.

tabulated value = 5134 + 0.28 PP + 0.03 PC1 =

0.28 PP = 0.28

steps:

calculate CT (Cumulative Total /

$CT = \frac{(\sum x_{TO})^2}{n_T}$

$n_T = \text{Total no. of tables} = \left[\frac{\sum (x_i^2)}{n} \right] - \frac{\sum x_i^2}{n} = 122$

$CT = \frac{(1000)^2}{30}$

$CT = 33333.33$

step 4: 0.03 PC1 = 0.28

SS - sum of squares:

$SS_{\text{Total}} = \sum x_{TO}^2 - CT$

= 35364 - 33333.33

$SS_{\text{Total}} = 2030.67$

0.28 PP = 0.28

step 5: calculate SS_t .

$$SS_t = \left(\frac{(\sum x_1)^2}{n_1} \right) + \left(\frac{(\sum x_2)^2}{n_2} \right) + \left(\frac{(\sum x_3)^2}{n_3} \right) - CT$$

$$= \frac{129600}{10} + \frac{99856}{10} + \frac{104976}{10} - 33333.33$$

$$= 12960 + 9985.6 + 10497.6 - 33333.33$$

$$SS_t = 129.86$$

step 6: calculate SS_e .

Individual SS .

$$SS_1 = \sum x_1^2 - \left[\frac{(\sum x_1)^2}{n_1} \right]$$

$$= 13252 - \left(\frac{129600}{10} \right)$$

$$SS_1 = 292$$

$$SS_e = SS_1 + SS_2 + SS_3$$

$$= 292 + 1146.4 + 482.4$$

$$SS_2 = \sum x_2^2 - \left(\frac{(\sum x_2)^2}{n_2} \right)$$

$$= 11132 - \left(\frac{99856}{10} \right)$$

$$SS_2 = 1146.4$$

$$SS_3 = \sum x_3^2 - \left(\frac{(\sum x_3)^2}{n_3} \right)$$

$$= 10980 - \left(\frac{104976}{10} \right)$$

$$SS_3 = 482.4$$

step 7: calculate $sse \rightarrow$ cross check sse .

$$SS_{\text{tot}} = SS_t + sse$$

$$sse = SS_{\text{tot}} - SS_t$$

$$= 2030.67 - 109.87$$

$$\boxed{sse = 1920.8}$$

step 8: determine no. of degree of freedom for each ss

$$df \text{ for } SS_{\text{tot}} = n_{\text{to}} - 1 = 30 - 1 = 29 //$$

$$df \text{ for } SS_t = a - 1 = 3 - 1 = 2 //$$

$$df \text{ for } sse = n_{\text{to}} - a = 30 - 3 = 27 //$$

step 9: estimate Mean of the square (MS). Note

$$MS_t = \frac{SS_t}{df_t}$$

$$= \frac{109.86}{2}$$

$$\boxed{MS_t = 54.935}$$

$$MS_e = \frac{sse}{df_e}$$

$$= \frac{1920.8}{27}$$

$$\boxed{MS_e = 71.14}$$

calculate for
variable
available
for the
problem

step 10: compute F (F statistics) $\frac{MS_{\text{(Variable)}}}{MS_{\text{error}}}$

$$F = \frac{MS_t}{MS_e} \rightarrow \text{Note}$$

$$= \frac{54.935}{71.14}$$

$$\boxed{F = 0.772}$$

$\rightarrow$ calculated F factor value.

Comparing calculated F factor with critical F value

Critical value of F factor taken from the value of

5% $\rightarrow 0.05$.

Finally

$$F_{8, 101} - F_{d, 0.05} =$$

$$8.05 P_1 = 922$$

step II : Construct ANOVA table $\otimes$ $\rightarrow$ Final answer table
should write as conclusion.

source of variation

SS, DF, MS, F, P

Treatment 109.87 2 54.935 0.772 > 0.05

Error

1920.8

27

71.14

Total

2030.67

29

Compare F value with critical value $F_{2, 27} = 3.95$.

There is no effect of different fertilizer doses on the
fresh weight of plants ($F_{2, 27} = 0.772, p > 0.05$).

Note:

Indirectly accepting the Null hypothesis and rejecting
the alternate hypothesis.

Randomised ^{Block} Completely Randomised Design
(RCBD).

Most widely used	2 blocks	1 member
* Blocks made	III	II
		II
		either each block

- * Blocks made
- * Treatments assigned at random within each block
- * complete randomization within a block
- * Each block with a separate randomization

Experiment : RCBD.

Total samples : 24

Blocks = $\frac{\text{samples}}{\text{treatment}} = \frac{24}{8} = 3 \rightarrow$ unavoidable variation.

Exp. error.

$$\text{Variability}_{\text{Total}} = \text{Variability}_t + \text{Variability}_B + \text{variability}_e.$$

$$\Delta F_{\text{P2POL}} = \frac{\gamma(\Delta \Gamma_{\text{POL}})}{8 \times 8} = \frac{\gamma(\Delta \Gamma_{\text{POL}})}{4.0 \times 10^7} = 70$$

Fresh weight (g) of plants under different doses of fertilizer (treatment A-H).

Treatment	Blocks			Treatment total	Treatment Mean
	I	II	III		
A	80.85	88.01	85.10	103.96	84.65
B	80.24	28.43	85.93	94.60	81.53
C	30.94	31.64	84.95	97.53	32.51
D	29.89	29.12	36.75	95.76	31.92
E	21.52	24.07	20.76	66.35	22.12
F	25.38	32.14	32.19	89.71	29.90
G	22.89	19.66	26.92	69.47	23.16
H	29.44	24.95	37.99	92.38	30.79
Block total	221.15	228.02	260.59	709.76	29.57

$$SS_{\text{total}} = SS_T + SS_B + SS_e$$

CT → cumulative total of correction term.

$$CT = \frac{(\sum x_{\text{total}})^2}{n_b \times n_t} = \frac{(709.76)^2}{3 \times 8} = 20989.96$$

SS_b → calculate for 3. variants.

$$SS_b = \left(\frac{(\sum x_1)^2}{n_1} \right) + \left(\frac{(\sum x_2)^2}{n_2} \right) + \left(\frac{(\sum x_3)^2}{n_3} \right) - CT +$$

$$\left(\frac{(\sum x_4)^2}{n_4} \right) + \left(\frac{(\sum x_5)^2}{n_5} \right) + \left(\frac{(\sum x_6)^2}{n_6} \right) + \left(\frac{(\sum x_7)^2}{n_7} \right) + \left(\frac{(\sum x_8)^2}{n_8} \right)$$

$$= \frac{103.96^2}{3} + \frac{94.60^2}{3} + \frac{97.53^2}{3} + \frac{95.76^2}{3} + \frac{66.35^2}{3} + \frac{89.71^2}{3} + \frac{69.47^2}{3} + \frac{92.38^2}{3} - 20989.96$$

$$= 3602.56 + \dots$$

$$= 21414.424 - 20989.96$$

$$SS_b = 426.46$$

calculate SS_b for all 8 variants samples treatment.

$$SS_b = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} + \frac{(\sum x_3)^2}{n_3} + \frac{(\sum x_4)^2}{n_4} + \frac{(\sum x_5)^2}{n_5} + \frac{(\sum x_6)^2}{n_6} + \frac{(\sum x_7)^2}{n_7} + \frac{(\sum x_8)^2}{n_8} - CT.$$

calculate SS_b for 3 blocks. → each block has 8 treatment

∴ 8 samples.

$$SS_b = \frac{(\sum x_{b1})^2}{n_1} + \frac{(\sum x_{b2})^2}{n_2} + \frac{(\sum x_{b3})^2}{n_3} - CT.$$

$$= \frac{(221.15)^2}{8} + \frac{(228.02)^2}{8} + \frac{(260.59)^2}{8} - 20989.96$$

$$= 21100.94 - 20989.96$$

$$SS_b = 110.98$$

$$= 20989.96$$

calculate SS_{tot}

$$SS_{tot} = \sum (x_{tot})^2 - CT$$

$$Total I^2 = 6214.79$$

$$Total II^2 = 6723.45$$

$$Total III^2 = 8730.14$$

$$\sum x_{total}^2 = 6214.79 + 6723.45 + 8730.14$$

$$= 21668.38$$

$$SS_{tot} = 21668.38 - 20989.96$$

$$= 678.42$$

$$SS_{tot} = 678.412$$